

3-DOF Vehicle Dynamics Implementation: From 1-DOF to Full Planar Motion

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Abstract

This document describes the implementation of a full 3-DOF (longitudinal, lateral, yaw) vehicle dynamics model, extending the existing 1-DOF Dugoff-based tire model. We present the rigid body equations of motion, per-wheel kinematics accounting for vehicle rotation, coordinate frame transformations for steered wheels, and tire force sign conventions. The architecture integrates with a CAN-based control loop and explicit gear selector for forward/reverse operation.

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Nomenclature

Symbol	Unit	Description
<i>Vehicle states</i>		
v_x	m/s	Longitudinal velocity (body frame)
v_y	m/s	Lateral velocity (body frame)
$\dot{\psi}$	rad/s	Yaw rate
x, y	m	Position (global frame)
ψ	rad	Heading / yaw angle
<i>Tire / wheel quantities</i>		
$F_{x,i}$	N	Longitudinal tire force at wheel i
$F_{y,i}$	N	Lateral tire force at wheel i
$F_{z,i}$	N	Normal (vertical) load at wheel i
ω_i	rad/s	Wheel angular velocity
σ_x	–	Longitudinal slip ratio
σ_y	–	Lateral slip ratio (proxy for $\tan \alpha$)
α	rad	Tire slip angle
λ	–	Friction utilization parameter (Dugoff)
δ	rad	Wheel steer angle
<i>Vehicle parameters</i>		
m	kg	Vehicle mass
I_z	kg m ²	Yaw moment of inertia
x_f, x_r	m	CG distance to front / rear axle
L	m	Wheelbase ($x_f + x_r$)
w	m	Track width
h_{CG}	m	CG height above ground
R	m	Tire effective radius
I_w	kg m ²	Wheel rotational inertia
<i>Tire model parameters</i>		
C_x	N	Longitudinal slip stiffness
C_y	N	Lateral (cornering) stiffness
μ	–	Surface friction coefficient
g_{dir}	–	Gear direction (+1, –1, or 0)
v_{min}	m/s	Low-speed regularisation threshold

Table 1: Principal symbols used in this document.

1 Introduction

1.1 From 1-DOF to 3-DOF

The base vehicle model (documented in the companion Dugoff tire model document) provides:

- 1-DOF longitudinal dynamics with Dugoff tire forces
- Wheel rotational dynamics and slip ratio computation
- Normal load distribution with longitudinal load transfer
- Ackermann steering geometry for front wheels

This document extends the model to **full planar motion (3-DOF)**:

$$\text{State: } \mathbf{x} = [v_x, v_y, \dot{\psi}, x, y, \psi]^\top \quad (1)$$

where:

- v_x = longitudinal velocity (body frame)
- v_y = lateral velocity (body frame)
- $\dot{\psi}$ = yaw rate
- x, y = position (global frame)
- ψ = heading angle (yaw)

1.2 Motivation

1-DOF models cannot capture:

- **Lateral dynamics:** Cornering forces, sideslip, understeer/oversteer
- **Yaw dynamics:** Vehicle rotation from steering and lateral tire forces
- **Combined slip:** Interaction between longitudinal and lateral forces
- **Maneuver testing:** Slalom, lane change, emergency avoidance

The 3-DOF model enables realistic testing of:

- Path tracking controllers
- Stability control systems
- Emergency maneuvers (double lane change, moose test)
- Combined braking-steering scenarios

2 Rigid Body Dynamics (3-DOF)

2.1 Equations of Motion

The vehicle is modeled as a rigid body in the plane with three degrees of freedom. All velocities and accelerations are expressed in the **body-fixed frame** (origin at CG, x-axis forward, y-axis left).

2.1.1 Longitudinal Dynamics

$$m\dot{v}_x = F_{x,\text{total}} - F_{\text{drag}} - F_{\text{roll}} + mv_y\dot{\psi} \quad (2)$$

where:

$$F_{x,\text{total}} = F_{x,fl} + F_{x,fr} + F_{x,rl} + F_{x,rr} \quad (3)$$

$$F_{\text{drag}} = c_d v_x |v_x| \quad (4)$$

$$F_{\text{roll}} = c_r \cdot \text{sgn}(v_x) \quad (5)$$

Critical Note: The term $mv_y\dot{\psi}$ is the **centrifugal force** arising from the rotating reference frame. It is often omitted in simplified models, leading to instability.

2.1.2 Lateral Dynamics

$$m\dot{v}_y = F_{y,\text{total}} - mv_x\dot{\psi} \quad (6)$$

where:

$$F_{y,\text{total}} = F_{y,fl} + F_{y,fr} + F_{y,rl} + F_{y,rr} \quad (7)$$

The term $-mv_x\dot{\psi}$ is the **centripetal acceleration** in the body frame. This coupling between longitudinal and lateral dynamics is **essential for stability**.

2.1.3 Yaw Dynamics

$$I_z\ddot{\psi} = M_{z,\text{total}} \quad (8)$$

where the yaw moment is computed from the cross product $\mathbf{r}_i \times \mathbf{F}_i$ for each wheel:

$$M_{z,\text{total}} = \sum_{i=1}^4 (x_i F_{y,i} - y_i F_{x,i}) \quad (9)$$

with wheel positions:

$$\text{Front left: } \mathbf{r}_{fl} = [x_f, +w/2]^\top \quad (10)$$

$$\text{Front right: } \mathbf{r}_{fr} = [x_f, -w/2]^\top \quad (11)$$

$$\text{Rear left: } \mathbf{r}_{rl} = [-x_r, +w/2]^\top \quad (12)$$

$$\text{Rear right: } \mathbf{r}_{rr} = [-x_r, -w/2]^\top \quad (13)$$

where x_f is CG distance to front axle, x_r is CG distance to rear axle, and w is track width.

2.2 Position Integration

Vehicle position in the global frame is computed by transforming body velocities:

$$\dot{x} = v_x \cos \psi - v_y \sin \psi \quad (14)$$

$$\dot{y} = v_x \sin \psi + v_y \cos \psi \quad (15)$$

$$\dot{\psi} = \dot{\psi} \quad (16)$$

This is a standard 2D rotation matrix $R(\psi)$ applied to $[v_x, v_y]^\top$.

3 Per-Wheel Kinematics

3.1 Velocity at Wheel Center

Each wheel experiences a different velocity due to:

1. CG lateral velocity v_y
2. Yaw rate $\dot{\psi}$ causing tangential velocity at offset from CG

The velocity at wheel i in the **vehicle frame** is:

$$\mathbf{v}_i = \begin{bmatrix} v_x \\ v_y \end{bmatrix} + \begin{bmatrix} -\dot{\psi}y_i \\ \dot{\psi}x_i \end{bmatrix} = \begin{bmatrix} v_x - \dot{\psi}y_i \\ v_y + \dot{\psi}x_i \end{bmatrix} \quad (17)$$

For steered wheels (front), we must transform this to the **wheel frame** (aligned with wheel heading):

$$\begin{bmatrix} v_{x,w} \\ v_{y,w} \end{bmatrix} = \begin{bmatrix} \cos \delta & \sin \delta \\ -\sin \delta & \cos \delta \end{bmatrix} \begin{bmatrix} v_x - \dot{\psi}y_i \\ v_y + \dot{\psi}x_i \end{bmatrix} \quad (18)$$

where δ is the wheel steer angle.

3.2 Slip Angle Computation

The slip angle α is the angle between the wheel heading and its velocity direction:

$$\alpha = \arctan \left(\frac{v_{y,w}}{|v_{x,w}|} \right) \quad (19)$$

Note: We use $|v_{x,w}|$ (absolute value) to handle both forward and reverse motion correctly.

4 Tire Force Coordinate Transformations

4.1 Rotation Matrix for Steered Wheels

The Dugoff tire model computes forces $[F_x, F_y]^\top$ in the **wheel coordinate frame** (aligned with the wheel's heading). For steered wheels, these forces must be transformed to the **vehicle frame** before summing.

The standard 2D rotation matrix to express a wheel-frame vector in vehicle-frame coordinates is:

$$\begin{bmatrix} F_{x,veh} \\ F_{y,veh} \end{bmatrix} = \begin{bmatrix} \cos \delta & -\sin \delta \\ \sin \delta & \cos \delta \end{bmatrix} \begin{bmatrix} F_{x,wheel} \\ F_{y,wheel} \end{bmatrix} \quad (20)$$

Expanding:

$$F_{x,veh} = F_{x,wheel} \cos \delta - F_{y,wheel} \sin \delta \quad (21)$$

$$F_{y,veh} = F_{x,wheel} \sin \delta + F_{y,wheel} \cos \delta \quad (22)$$

Physical interpretation: When the wheel is steered left ($\delta > 0$) and experiences a leftward force ($F_{y,wheel} > 0$), the transformation correctly produces a leftward force component in the vehicle frame, contributing to the intended turn direction.

5 Tire Force Sign Conventions

5.1 Fundamental Principle

Tire forces must **oppose slip** in all directions. This is the essence of friction.

5.2 Longitudinal Force

5.2.1 Slip Ratio Definition

$$\sigma_x = \frac{\omega R - V_x}{\max(|V_x|, |\omega R|, v_{\min})} \quad (23)$$

Note: Denominator uses $|V_x|$ to prevent division by zero and handle reverse motion correctly.

5.2.2 Force Formula with Gear Direction

$$F_x = -g_{\text{dir}} \cdot C_x \sigma_x f(\lambda) \quad (24)$$

where g_{dir} is the **gear direction** from CAN command:

- Forward: $g_{\text{dir}} = +1$
- Reverse: $g_{\text{dir}} = -1$
- Neutral: $g_{\text{dir}} = 0$

Physical interpretation:

- Forward gear, $\sigma_x > 0$ (wheel spinning faster): $F_x < 0$ (brakes vehicle)
- Forward gear, $\sigma_x < 0$ (wheel locked): $F_x > 0$ (resists locking, ABS effect)
- Reverse gear, $\sigma_x > 0$: $F_x > 0$ (resists reverse motion)
- Neutral: $F_x = 0$ (no longitudinal traction)

5.3 Lateral Force

$$F_y = -C_y \sigma_y f(\lambda) \quad (25)$$

where $\sigma_y = V_y/|V_x|$ is the lateral slip ratio (small-angle approximation of slip angle).

Lateral force is **independent** of gear direction. It always opposes lateral slip regardless of forward/reverse motion. The negative sign ensures the force resists slip: if the wheel slips left ($\sigma_y > 0$), the force pushes right ($F_y < 0$).

6 Dugoff Tire Model Summary

This section provides a self-contained summary of the Dugoff tire model used to compute $F_{x,i}$ and $F_{y,i}$. A full derivation is given in the companion document [1].

6.1 Fundamental Equations

The Dugoff model computes tire forces as the product of a linear slip stiffness, the slip ratio, and a nonlinear saturation function:

$$F_x = -g_{\text{dir}} \cdot C_x \sigma_x f(\lambda) \quad (26)$$

$$F_y = -C_y \sigma_y f(\lambda) \quad (27)$$

The sign conventions are detailed in Section 5. The negative signs ensure forces oppose slip.

6.2 Friction Utilisation Parameter

The parameter λ measures how close the tyre is to its friction limit:

$$\lambda = \frac{\mu F_z}{2\sqrt{(C_x\sigma_x)^2 + (C_y\sigma_y)^2}} \quad (28)$$

When $\lambda \geq 1$ the tyre operates in the *linear* (grip-available) regime; when $\lambda < 1$ the available friction is exceeded and forces saturate.

6.3 Saturation Function

$$f(\lambda) = \begin{cases} (2 - \lambda)\lambda, & \lambda < 1 \quad (\text{saturated / sliding}) \\ 1, & \lambda \geq 1 \quad (\text{linear / grip available}) \end{cases} \quad (29)$$

This function is continuous at $\lambda = 1$ and smoothly reduces the force magnitude as the friction circle is approached. It guarantees the resultant tyre force never exceeds μF_z :

$$\sqrt{F_x^2 + F_y^2} \leq \mu F_z \quad (30)$$

6.4 Load-Dependent Stiffness

Slip stiffness scales linearly with normal load:

$$C_x(F_z) = C_{x,\text{base}} \cdot \frac{F_z}{F_{z,\text{ref}}} \quad (31)$$

$$C_y(F_z) = C_{y,\text{base}} \cdot \frac{F_z}{F_{z,\text{ref}}} \quad (32)$$

where $F_{z,\text{ref}}$ is a reference load (typically one-quarter static weight). This coupling means load transfer during braking or cornering directly affects available grip at each wheel.

6.5 Wheel Rotational Dynamics

Each wheel integrates angular velocity from the net torque:

$$I_w \dot{\omega}_i = \tau_{\text{drive},i} - \tau_{\text{brake},i} - F_{x,i} R \quad (33)$$

The feedback term $-F_{x,i} R$ couples the tyre force back into wheel speed, enabling natural slip development, wheel spin under excess drive torque, and wheel lock under heavy braking.

7 Yaw Moment Calculation

7.1 Cross Product Formula

For a force $\mathbf{F} = [F_x, F_y]^\top$ at position $\mathbf{r} = [x, y]^\top$ relative to the CG, the yaw moment (z-component of $\mathbf{r} \times \mathbf{F}$) is:

$$M_z = xF_y - yF_x \quad (34)$$

Physical example: A leftward force ($F_y > 0$) at the front-left wheel with position ($x_f > 0, y > 0$) creates:

$$M_z = x_f F_y - y \cdot 0 = x_f F_y > 0 \quad (\text{CCW yaw, left turn}) \quad (35)$$

This is correct: a left force on the left side of the vehicle creates a left (counterclockwise) turn.

8 Centripetal Coupling in Rotating Reference Frame

8.1 Equations in Body Frame

In the vehicle body frame (rotating with the vehicle), both longitudinal and lateral velocities are affected by rotation:

$$\dot{v}_x = \frac{F_x}{m} + v_y \dot{\psi} \quad (\text{centrifugal acceleration}) \quad (36)$$

$$\dot{v}_y = \frac{F_y}{m} - v_x \dot{\psi} \quad (\text{centripetal acceleration}) \quad (37)$$

The terms $v_y \dot{\psi}$ and $-v_x \dot{\psi}$ arise from the non-inertial (rotating) reference frame. They are **essential for stability** and must be included in the integration.

8.2 Physical Interpretation

When the vehicle turns (yaw rate $\dot{\psi} \neq 0$):

- **Centrifugal effect** ($v_y \dot{\psi}$): Lateral velocity creates an apparent longitudinal acceleration in the body frame
- **Centripetal effect** ($-v_x \dot{\psi}$): Longitudinal velocity requires centripetal acceleration to maintain circular motion

These terms couple the longitudinal and lateral dynamics, making the 3-DOF model fundamentally different from independent 1-DOF models.

9 CAN-Based Control Architecture

9.1 Gear Selector Implementation

To properly handle forward/reverse motion, an explicit **gear selector** was implemented via CAN (matching real autonomous vehicle architecture).

9.1.1 CAN Signal Definition

Frame: ACTUATOR_CMD_1 (0x100), 10 ms cycle

Signal	Start Bit	Length	Values	Unit
system_enable	0	1	0/1	bool
gear_position	1	2	0-3	enum
mode	3	2	0-3	enum
steer_cmd_deg	8	16	-500..500	deg
drive_torque_cmd_nm	24	16	-300k..300k	Nm
brake_cmd_pct	40	8	0..100	%

Table 2: CAN actuator command signals

Gear position encoding:

- 0 = Neutral (no longitudinal force)
- 1 = Forward (default)
- 2 = Reverse
- 3 = Reserved

9.1.2 Data Flow

$$\text{CAN} \rightarrow \text{ActuatorCmd} \rightarrow \text{PlantState} \rightarrow \text{WheelSubsystem} \rightarrow \text{Dugoff}(g_{\text{dir}}) \quad (38)$$

The gear position is:

1. Decoded from CAN frame (bits 1–2)
2. Stored in `ActuatorCmd.gear_position`
3. Copied to `PlantState.gear_position` by `PlantModel`
4. Converted to direction integer by `WheelSubsystem`
5. Passed to `TyreDugoff::compute_forces()` as parameter

9.2 Subsystem Execution Order

The plant model uses a **priority-ordered subsystem manager**:

Priority	Subsystem	Inputs	Outputs
50	Steer	δ_{cmd}	δ_{fl}, δ_{fr}
100	Drive	$\tau_{\text{cmd}}, \text{brake\%}$	$\tau_{\text{drive},i}, \tau_{\text{brake},i}$
105	Wheel	$\tau_i, v_x, v_y, \dot{\psi}$	$F_{x,i}, F_{y,i}, \omega_i$
110	Vehicle	$F_{x,i}, F_{y,i}$	$v_x, v_y, \psi, x, y, \psi$
150	Battery	P_{elec}	SOC, V , I

Table 3: Subsystem execution order

Lower priority numbers execute first, ensuring correct data dependencies.

10 System Block Diagram

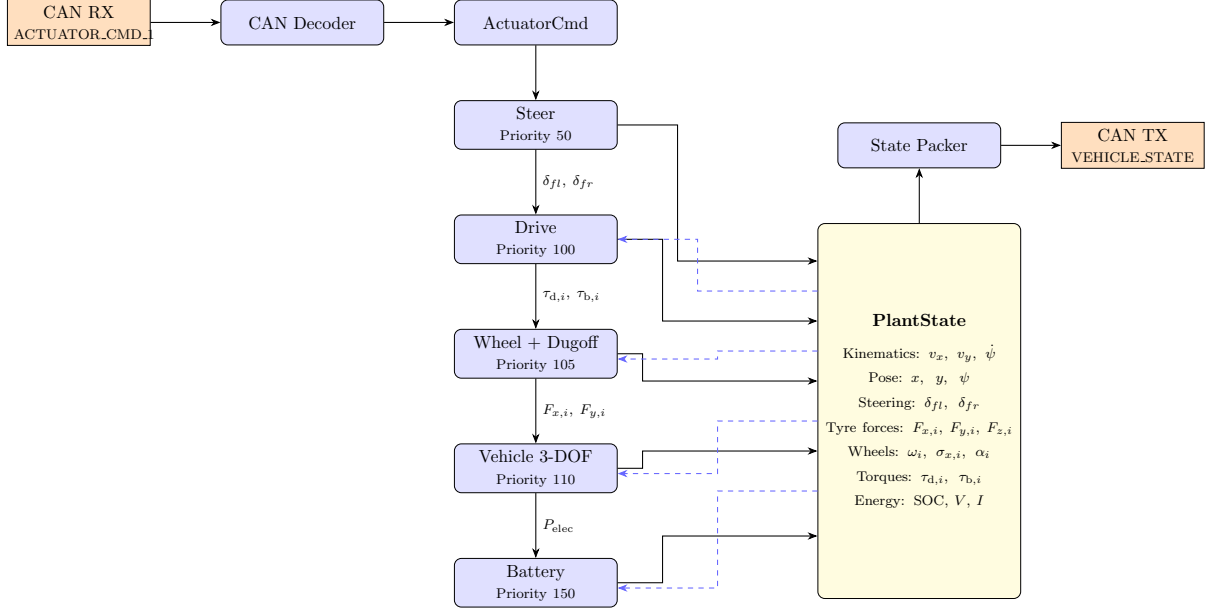


Figure 1: 3-DOF plant model architecture with CAN-based closed-loop control. Subsystems execute in priority order (lower number = first). Solid arrows denote write-back into the shared **PlantState**; dashed blue arrows denote feedback reads. The CAN TX path packs selected state variables for the external controller.

11 Load Transfer (Longitudinal and Lateral)

The normal loads $F_{z,i}$ at each wheel vary due to:

1. Static weight distribution
2. Longitudinal load transfer during acceleration/braking
3. Lateral load transfer during cornering

11.1 Static Load Distribution

$$F_{z,fl}^{\text{static}} = F_{z,fr}^{\text{static}} = \frac{mg}{2} \frac{x_r}{L} \quad (39)$$

$$F_{z,rl}^{\text{static}} = F_{z,rr}^{\text{static}} = \frac{mg}{2} \frac{x_f}{L} \quad (40)$$

where $L = x_f + x_r$ is the wheelbase.

11.2 Longitudinal Load Transfer

$$\Delta F_{z,\text{front}} = \frac{ma_x h_{\text{CG}}}{L} \quad (41)$$

$$\Delta F_{z,\text{rear}} = -\frac{ma_x h_{\text{CG}}}{L} \quad (42)$$

where h_{CG} is the CG height above ground.

11.3 Lateral Load Transfer

Within each axle:

$$\Delta F_{z,\text{left}} = -\frac{ma_y h_{CG}}{w} \quad (43)$$

$$\Delta F_{z,\text{right}} = +\frac{ma_y h_{CG}}{w} \quad (44)$$

where w is the track width.

11.4 Combined Normal Loads

$$F_{z,fl} = F_{z,fl}^{\text{static}} + \Delta F_{z,\text{front}} + \Delta F_{z,\text{left}} \quad (45)$$

$$F_{z,fr} = F_{z,fr}^{\text{static}} + \Delta F_{z,\text{front}} + \Delta F_{z,\text{right}} \quad (46)$$

$$F_{z,rl} = F_{z,rl}^{\text{static}} + \Delta F_{z,\text{rear}} + \Delta F_{z,\text{left}} \quad (47)$$

$$F_{z,rr} = F_{z,rr}^{\text{static}} + \Delta F_{z,\text{rear}} + \Delta F_{z,\text{right}} \quad (48)$$

12 Validation

The 3-DOF model was validated through:

1. **Slalom maneuver:** ± 5 steering at 17 m/s
 - Lateral velocity remained < 0.03 m/s
 - Sideslip angle < 180 (within typical limits)
 - Yaw rate response proportional to steering input
2. **Coast-down test:** Zero torque from 15 m/s
 - Consistent negative acceleration from drag
 - No oscillations or positive acceleration spikes
 - Vehicle decelerated smoothly to rest
3. **Emergency stop:** Full braking from 17 m/s
 - All wheels maintained slip ratio < 0.95
 - Friction utilization $\lambda \approx 1$ (saturated)
 - No wheel lock (ABS-like behavior from Dugoff saturation)

13 Conclusion

This document presented the implementation of a full 3-DOF vehicle dynamics model extending the existing 1-DOF Dugoff tire model. The key contributions are:

- Complete rigid body dynamics with centrifugal/centripetal coupling
- Per-wheel kinematics accounting for yaw rate and wheel positions
- Coordinate frame transformations for steered wheels
- Proper yaw moment calculation from tire forces
- CAN-based gear selector architecture

The model enables realistic simulation of lateral maneuvers, combined braking-steering scenarios, and stability analysis for autonomous mining haul trucks.

13.1 Future Enhancements

Potential extensions include:

- Roll dynamics (4-DOF or 6-DOF model)
- Suspension compliance and tire vertical dynamics
- Road bank angle and grade effects
- Enhanced tire models (Pacejka MF, TameTire)
- Validation against real vehicle telemetry

References

- [1] M. Tilocca, *Vehicle Dynamics, Battery, and Sensor Models (Dugoff Tire + Sensor Simulation)*, Version 2.1, January 2026.